

**Project Presentation of CO-OP Project at Industry (Module-2) (22CS421)**

On

# Sponsored Search and User Attention

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# Objective

## **Understand**

Examine how sponsored search results influence user attention, trust, and decision-making behavior during online searches.

## **Study**

Investigate the relationship between advertisement relevance, click behavior, and user trust formation.

## **Compare**

Analyze the differences between organic and sponsored links within search engine result pages (SERPs).

## **Explore**

Understand how cross-platform advertising contributes to organic visibility and search ranking authority.

# Introduction

- Search engines display two types of results:
  - Organic Results (merit-based)
  - Sponsored Results (paid advertisements)
- Sponsored links are a major revenue source for search engines.
- User trust and relevance strongly influence interaction with sponsored ads.
- Sponsored search impacts both decision-making and perception.

# Introduction

## Organic Ranking

- Merit-based Ranking
- Higher Trust
- SEO Dependent

## Sponsored Ranking

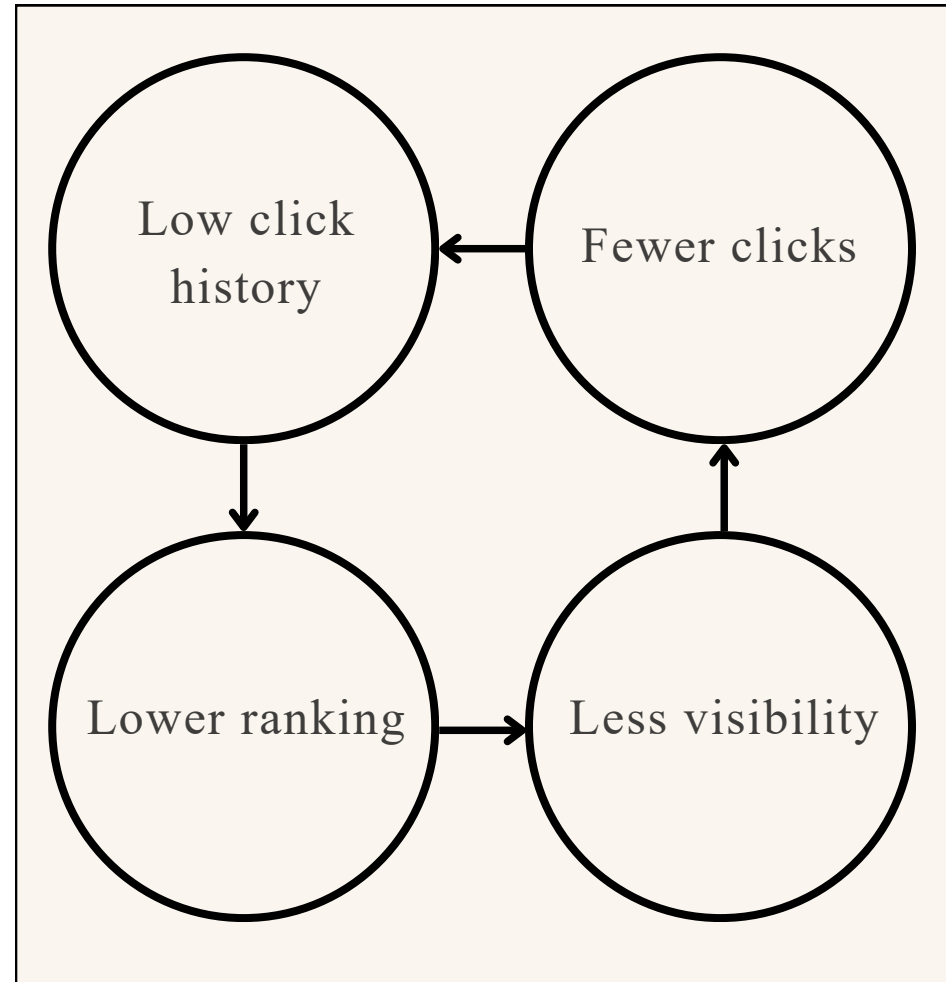
- Paid advertisements
- Revenue Generating
- Bid dependent

# Problem Statement

- Click-based ranking systems prioritize ads with higher click history.
- Highly relevant ads may remain unseen due to low CTR.
- Users often trust organic results more than sponsored results.
- Transparency and fairness in ranking remain challenges.



## Problem Statement



# Methodology

**1****Literature Review**

Existing literature and prior research on sponsored search, user trust, and search behavior were examined.

**2****Conceptual Analysis**

Conceptual analysis was performed to identify relationships between relevance, attention, and sponsored results.

**3****Python simulation**

Python-based simulations were developed using datasets to model sponsored search ranking behavior.

**4****Result Validation**

Simulation outcomes were compared with theoretical findings to support the research conclusions.

# Python Validation

## **Python Simulation**

Python simulations were used to support and verify the research findings through computational modeling.

## **Synthetic Dataset Creation**

Synthetic datasets were generated to model sponsored search behavior and ranking patterns.

## **Model Implementation**

A Random Forest model was implemented to analyze relevance prediction using simulated search data.

## **Ranking Comparison & Validation**

CTR-based ranking was compared with relevance-based ranking to validate the research findings.



# Key Findings

- High click-through rate does not always indicate high relevance.
- Many relevant advertisements may receive very few clicks.
- Sponsored links influence user trust and decision-making.
- Organic results are generally viewed as more trustworthy.
- Cross-platform visibility can improve organic ranking performance.

# Conclusion

## **Behavioral Studies**

Real-world behavioral studies can provide deeper insights into user interaction and search patterns.

## **Mobile Search Analysis**

Mobile search interaction can be explored further to understand user engagement across devices.

## **AI Search Environments**

AI-generated search environments can be analyzed to study evolving ranking dynamics.

## **Trust & Ranking Models**

Cross-cultural research and improved ranking models can enhance trust evaluation and reduce advertiser bias.

# Future Scope

- 1 Behavioral Studies**  
Future work can include real-world behavioral studies to understand user interaction patterns.
- 2 Mobile Search Interaction**  
Mobile search interaction can be explored further to examine engagement across devices.
- 3 AI Search Environments**  
AI-generated search environments may be analyzed to study evolving search behavior.
- 4 Trust & Ranking Models**  
Cross-cultural studies and improved ranking models can strengthen trust evaluation and reduce advertiser bias.



Thank You